Real Estate Price Prediction

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ABSTRACT

This research explores an enhanced machine learning framework for predicting real estate prices in Mumbai, a city characterized by significant spatial, economic, and environmental diversity. Traditional valuation approaches rely solely on structural attributes such as area, BHK count, and locality, which limits their ability to capture the deeper socio-economic and environmental influences shaping housing markets. To address this challenge, the present study integrates structural property data with geo-spatial indicators including NDVI, NDBI, night-light intensity, and geographic coordinates. Twelve machine learning algorithms were evaluated, and the Gradient Boosting Regressor emerged as the strongest model, achieving an R^2 score of 0.8560 with stable generalization across 228 regions. The results demonstrate substantial accuracy improvements when satellite and spatial features are incorporated, establishing the effectiveness of a multimodal approach for metropolitan real-estate prediction.

1. INTRODUCTION

Real estate valuation is crucial in guiding investment decisions, urban planning, and policy formulation. In a highly dynamic and densely populated city like Mumbai, property prices are influenced by a wide range of factors that extend beyond conventional structural attributes. The city's heterogeneous development patterns, variations in greenery, infrastructure availability, built-up density, and economic activity create significant micro-level price differences across neighbourhoods. Traditional statistical models fail to capture these complex interactions because they primarily analyze linear relationships. Machine learning techniques provide stronger predictive capability but remain limited when used without spatial and environmental context. This study seeks to bridge that gap by integrating geo-spatial intelligence with machine learning, thereby developing a more accurate and comprehensive prediction model.

2. LITERATURE REVIEW

2.1 Traditional Statistical Approaches

Early real-estate valuation techniques relied heavily on hedonic pricing models and linear regression. These methods focused primarily on structural property features such as BHK count, carpet area, and furnishing details, assuming that pricing relationships were linear and stable across regions. While their interpretability made them easy to deploy, several studies revealed their significant limitations in metropolitan environments characterized by spatial diversity and irregular development patterns. In cities like Mumbai, where neighbourhood-level socio-economic contrasts are stark, such models performed poorly because they were unable to capture non-linear interactions between environmental factors, infrastructure conditions, and market fluctuations. Research using regression-based approaches consistently reported modest accuracy, highlighting their inability to represent the true complexity of urban real-estate behaviour.

2.2 Machine Learning and Ensemble-Based Models

As the limitations of classical models became evident, researchers shifted toward machine learning algorithms capable of capturing non-linear patterns. Decision-tree-based methods, including Random Forest, Gradient Boosting, and XGBoost, demonstrated superior predictive strength due to their ability to learn intricate feature interactions and adapt to heterogeneous datasets. Comparative studies showed that boosting-based models frequently outperformed linear regressors on benchmark datasets. However, despite these improvements, many studies continued relying almost entirely on structured tabular data. Without spatial or environmental information, even the strongest ensemble models struggled to generalize effectively across diverse metropolitan regions. This revealed a major methodological gap: machine learning models alone cannot fully explain property valuation trends unless enriched with meaningful contextual features.

2.3 Integration of Satellite Imagery and Geo-Spatial Intelligence

Recent advancements in geo-spatial analytics have encouraged researchers to incorporate satellite imagery, vegetation indices, built-up density metrics, and night-time light data into housing price prediction frameworks. Multimodal approaches—such as those using CNNs on satellite and street-view imagery—have successfully captured neighbourhood characteristics, greenery distribution, building quality, and infrastructure development. Studies in remote sensing have established strong correlations between NDVI, NDBI, night-light intensity, and economic activity, suggesting that environmental quality and urban density directly influence property valuation. Still, many such studies were conducted in cities outside India or used limited datasets, reducing their general applicability in complex, high-density markets like Mumbai. Although multimodal imaging approaches achieved high accuracy, they often lacked a unified framework that merges environmental indicators with structured real-estate data.

2.4 Gaps in Current Research and Need for a Mumbai-Focused Framework

Despite progress in multimodal modelling, existing research falls short in addressing the unique challenges of Indian megacities. Mumbai's extreme population density, rapid infrastructure evolution, environmental variability, and socio-economic fragmentation create complex real-estate dynamics that most global models do not account for. Few studies have attempted to integrate structural, spatial, and environmental datasets into a single predictive pipeline tailored specifically for Mumbai. Many prior models lack satellite-derived indicators, ignore intra-city variations, or fail to represent region-specific housing behaviour. This research directly responds to those gaps by incorporating NDVI, NDBI, night-light intensity, and geo-coordinates into a comprehensive machine learning framework designed for Mumbai's heterogeneous market. The unified multimodal approach allows the model to capture intricate spatial patterns, producing significantly improved accuracy across 228 neighbourhoods and establishing a strong foundation for region-aware valuation systems.

3. METHODOLOGY

3.1 Dataset Construction and Integration

The methodology began with the development of a comprehensive dataset that combined property-level information with environmental and geo-spatial indicators. The foundation of the dataset was the Mumbai Housing Prices dataset, which contains essential structural attributes such as price, carpet area, BHK configuration, furnishing details, and property type. To enrich this dataset with contextual intelligence, multiple geo-spatial variables were extracted using Google Earth Engine. The Normalized Difference Vegetation Index (NDVI) served as a proxy for greenery and environmental quality, the Normalized Difference Built-up Index (NDBI) reflected urbanization density, and night-light intensity represented economic activity and infrastructure concentration. Additionally, latitude and longitude values were assigned to 228 regions using the Geopy library, enabling the model to identify spatial clusters and neighbourhood-specific behavioural patterns. The integration of these diverse data sources resulted in a multimodal dataset comprising more than 55,000 records and 14 enriched features, offering a detailed representation of Mumbai's heterogeneous real-estate landscape.

3.2 Data Preprocessing and Feature Engineering

To ensure reliability and consistency, the collected dataset underwent a rigorous preprocessing pipeline. Outlier removal was performed to eliminate unrealistic or extreme values, particularly with respect to price-per-square-foot distributions, which could distort the learning process. Missing values were imputed using appropriate statistical strategies: numerical variables were filled using medians to minimize skew, while categorical attributes were imputed using the most frequent category. Location information—originally containing spelling variations, duplicate entries, and inconsistent formatting—was standardized to maintain uniformity across all records. Feature engineering played a critical role in enhancing model performance, with the introduction of price_per_sqft as an engineered metric that normalized price variations across differently sized properties. The merging of structural, environmental, and spatial components produced a unified dataset suitable for advanced machine learning workflows.

3.3 Model Development and Training Strategy

Following preprocessing, a structured model development pipeline was implemented. Multiple categories of models were explored to ensure comprehensive benchmarking across different algorithmic families. The initial phase involved training baseline models such as Linear Regression, Ridge, Lasso, and Decision Trees, which provided foundational performance benchmarks. Subsequently, more advanced machine learning techniques were introduced, including Random Forest, Gradient Boosting Regressor, XGBoost, LightGBM, CatBoost, and Stochastic Gradient Descent Regression. These models were selected for their ability to capture non-linear dependencies and complex interactions between structural and geo-spatial inputs. In addition to individual learners, ensemble techniques such as Voting Regressor and Stacking Regressor were also developed to examine the benefits of combining multiple strong learners. This multi-stage modelling strategy allowed the study to systematically compare simple, intermediate, and highly optimized models under a unified experimental framework.

3.4 Model Evaluation Framework

The performance of each machine learning model was assessed using a consistent evaluation protocol, incorporating the metrics Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and the Coefficient of Determination (R^2). These metrics provided a balanced perspective on both the magnitude of prediction errors and the overall explanatory power of each model. A standard train–test split ensured fair evaluation, with selective hyperparameter tuning applied to advanced models to enhance their predictive accuracy. The combination of multimodal data, diverse model architectures, and robust evaluation methods enabled the identification of the most effective predictive framework. Ultimately, the Gradient Boosting Regressor was chosen as the final model due to its strong performance, minimal overfitting, and computational efficiency, demonstrating high generalization capability across Mumbai’s 228 regions.

4. RESULTS

4.1 Comparative Performance of Machine Learning Models

The evaluation of twelve machine learning models revealed significant differences in their ability to predict real estate prices within Mumbai's spatially complex environment. Traditional regression models such as Linear Regression, Ridge, and Lasso exhibited weak performance, consistently yielding R^2 values between 0.76 and 0.77. These models were unable to capture the complex non-linear dependencies and spatial variations influencing property prices across the 228 regions. Decision Tree models showed moderate improvement, but their tendency to overfit made them unreliable when generalizing to unseen data. In contrast, ensemble-based approaches demonstrated superior predictive strength, particularly those leveraging boosting techniques. Models such as XGBoost, LightGBM, and CatBoost delivered notably higher R^2 scores, reflecting their ability to learn intricate relationships embedded within multimodal features. Nevertheless, the Gradient Boosting Regressor emerged as the most stable and balanced model, achieving an R^2 score of 0.8560 on the test set and exhibiting highly consistent performance across training and validation datasets.

4.2 Impact of Geo-Spatial and Satellite-Derived Features

A critical aspect of the experimental evaluation involved assessing the contribution of geo-spatial and satellite-derived indicators such as NDVI, NDBI, night-light intensity, and latitude–longitude coordinates. A comparative analysis was conducted between two versions of each model—one trained solely on structural housing attributes and the other using the multimodal enriched dataset. The results showed a substantial improvement in accuracy when spatial and environmental features were incorporated. Models trained without geo-spatial indicators typically achieved R^2 values between 0.70 and 0.78, indicating limited ability to differentiate pricing behaviour across diverse neighbourhoods. However, upon integrating NDVI, NDBI, and night-light features, the performance jumped to the range of 0.82–0.86, demonstrating a 10–20% improvement in predictive accuracy. This enhancement underscores that environmental quality, built-up density, and night-time economic activity are vital factors influencing housing valuation, particularly in a densely populated and economically varied city like Mumbai.

4.3 Region-Wise Evaluation and Behavioural Trends

To better understand the spatial generalization of the Gradient Boosting model, region-wise performance was analyzed across Mumbai's distinct urban zones. In South Mumbai, characterized by dense built-up structures, premium amenities, and strong economic activity, the model accurately captured high property valuations due to strong correlations between pricing patterns and elevated NDBI and night-light intensity levels. In the Western and Central suburbs, where development is mixed and greenery varies significantly between neighbourhoods, the model produced stable mid-range predictions consistent with observed market behaviour. Navi Mumbai, a region experiencing rapid infrastructural development, showed pricing trends strongly aligned with night-light intensity and built-up expansion indicators, both of which were effectively recognized by the model. Regions such as Thane, which exhibit higher NDVI values and more abundant green coverage, reflected relatively lower pricing trends compared to central metropolitan zones, demonstrating the model's ability to interpret environmental influences with accuracy. Overall, the consistent region-wise performance confirms that the multimodal approach effectively captures Mumbai's diverse housing market structure.

4.4 Error Distribution and Generalization Capability

The residual analysis further validated the model's robustness and generalization capability. Errors were approximately normally distributed across the dataset, indicating that the model did not exhibit systematic overestimation or underestimation tendencies. Only a small number of large residuals were recorded, primarily associated with atypical luxury properties or rare listings that deviated significantly from general pricing patterns. The narrow gap between the training R^2 (0.8772) and testing R^2 (0.8560) demonstrates minimal overfitting and confirms that the Gradient Boosting Regressor maintains predictive stability even when exposed to unseen data. Its strong performance across multiple regions, combined with consistent error behaviour, establishes it as a reliable and scalable model for real-estate valuation in complex metropolitan markets.

5. DISCUSSION

The results of this study highlight the limitations of traditional attribute-based models in capturing the complex and spatially diverse nature of Mumbai's real-estate market. Structural features alone were insufficient to represent the strong influence of environmental and infrastructural factors that vary significantly across regions. The substantial improvement in prediction accuracy after incorporating geo-spatial and satellite-derived indicators such as NDVI, NDBI, night-light intensity, and precise geographic coordinates demonstrates the critical role of multimodal feature integration in real-estate valuation. These features allowed the models to recognize greenery distribution, urban density, and economic activity, enabling more context-aware predictions that reflect real market behaviour.

Among all machine learning techniques evaluated, boosting algorithms—particularly the Gradient Boosting Regressor—proved most effective due to their ability to model non-linear interactions and learn from residual errors iteratively. This model offered a strong balance between accuracy, generalization, and computational efficiency, outperforming both traditional regression methods and more resource-intensive ensemble architectures. The findings underscore the importance of geo-spatial intelligence in future real-estate prediction frameworks and suggest that multimodal modelling can significantly enhance valuation accuracy in metropolitan environments.

6. CONCLUSION

This study demonstrates that integrating structural, environmental, and geo-spatial intelligence significantly enhances real-estate price prediction in heterogeneous metropolitan contexts. The Gradient Boosting Regressor proved to be the most reliable model, achieving high predictive accuracy and stable generalization across diverse regions of Mumbai. The framework developed in this research provides a practical foundation for real-estate analytics applications and can be adapted for other cities with appropriate spatial data. Future work may incorporate temporal analysis, deep learning-based image processing, and additional socio-economic indicators to further refine predictive capability.

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